Frontend Application Tier

Core Design System & Layout
Tailwind CSS • Lucide Icons

Auth State & Route Guards
AuthContext • ProtectedRoutes

Provider Management Suite
Dashboard • Services • Hours • Time-Off

Customer Booking Suite
Slot Picker • Real-time Grid • 409 Modal • Rescheduler

Axios Central API Client
JWT Bearer Interceptors

HTTPS + JWT + Idempotency-Key

Security & Gateway Boundary

Reverse Proxy / TLS Gateway

SimpleJWT Bearer Token Guard

Django CORS Security Filter

Unified Exception Handler
PostgreSQL violations → HTTP 409

DRF API Gateway Router
/api/v1/

API requests

Backend Domain Services

Core & Timezone Engine
IANA zoneinfo → UTC • DST-safe

Accounts Service
CustomUser • Customer • Provider • Admin

Providers & Services Engine
Durations • Buffers • Notice Limits

Dynamic Slot Calculation Engine
Time-off • Buffers • Active Bookings

Open slots in client/provider timezones

External & Notification Services

Atomic Booking Engine
transaction.atomic • Create • Reschedule • Cancel

Confirmed booking → reminders

Background Worker
run_reminders • SELECT FOR UPDATE SKIP LOCKED

AI Intelligence Suite
Concierge • Intake Brief • Waitlist Recovery

Mock Notification Service
Email/SMS • Exponential Retry Backoff

Notification dispatch

Email/SMS delivery

Database Tier

Tables: users • providers • services • working_hours

Tables: time_off • bookings • booking_events • reminders

Zero Double-Booking Invariant
GiST EXCLUDE provider_id + tstzrange overlap

Reminders Queue
status='pending' • scheduled_for index

SKIP LOCKED reminder processing

OpenAI API
AI Concierge • Pre-meeting Summaries

AI Concierge and summaries

Atomic booking write